Beyond Matching: Structural and Case Constraints in Russian Free Relatives[1]

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Introduction. Free relative clauses (FRs) are clauses without an overt noun head that have a distribution of the corresponding nominal constituent (Grimshaw 1977). Several analysis of FRs have been proposed, including the Head Hypothesis (Bresnan, Grimshaw 1978; Larson 1987, 1998), the COMP Hypothesis (Groos, van Riemsdijk 1981; Harbert 1983; Suñer 1983, 1984; Grosu 1994), and shared-structure analyses (van Riemsdijk 2006; Bergsma 2019), In many languages, FRs are subject to **case and categorial matching conditions** between the *wh*-expression at the left periphery of the FR and the syntactic position (the null head Citko 2004; Šimík 2018) in the main clause. Some of the proposed approaches to FRs analyze the relation between the null head and the *wh*-expression as *Agree* relation (Spyropolous 2011, Himmelreich 2017), while others propose PF deletion of a light head (Bergsma 2023). Russian FRs have been argued to obey case and categorial matching restrictions (Lyutikova 2015). However, the gradient nature of these restrictions and the factors affecting acceptability have not previously been investigated. The present study provides the first experimental investigation of Russian FRs.

Methodology. All experiments were conducted on the PCIbex Farm platform (Zehr, Schwarz 2018) using an acceptability judgment task (1–7 Likert scale). Alongside the experimental stimuli, each experiment included fillers containing embedded questions introduced by the same case and categorial form of the *wh*-expression used in the corresponding stimuli. The judgements were z-normalized to avoid the scale bias. Linear mixed-effect regression models were fitted to the data, followed by Tukey pairwise comparisons.

Case matching and case hierarchy. Experiments 1 and 2 investigate the acceptability of different case configurations of the null head and the *wh*-expression. While languages like German (Mewe 2019) show an effect of case hierarchy (Caha 2009), with configurations in which the null head bears a less marked case than the *wh*-phrase (i.e. a *harmonic configuration*) being more preferred, it has remained unclear whether the similar pattern holds in Russian. The independent variables in Experiment 1 were: 1) null head case (NOM/ACC), 2) *wh*-case (NOM/ACC), 3) position of the FR (postposition/preposition), in Experiment 2: 1) head-wh configuration (harmonic (ACC[DAT] vs. ACC[INS]) / non-harmonic (DAT[ACC], INS[ACC])), 2) position of the FR. The results are presented in Figures 1–2 (Notation X[Y] = case/category of the null head (X) [case/category of the *wh*-expression (Y)]), significant differences between conditions are marked with an asterisk. The example of ACC[NOM] mismatch is presented in (1). The results show that case mismatches significantly reduce acceptability, but harmonic case mismatches receive higher ratings than non-harmonic ones only when the FR is preposed.

Case syncretism. Experiment 3 explored the effect of ACC-GEN syncretism in the *wh*-expression *kto* 'who' on the acceptability of case mismatches. The results (Figure 3) show that mismatch does not incur an acceptability penalty when the two cases have the same morphological realization. One possible account is based on Impoverishment in Distributed Morphology (Halle, Marantz 1993): GEN → ACC / [+Animate]. The structural representation of GEN is reduced to the ACC representation, neutralizing the feature conflict. The fact that the effect is independent of linear position suggests that syncretism may affect case matching at an earlier stage of the derivation, potentially in narrow syntax.

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Categorial matching. Experiment 4 tested whether the null head and the *wh*-expression must match in syntactic category. Two types of mismatch were investigated: DP[PP]: the null head is a DP, while the *wh*-expression is contained in a PP; PP[DP]: the null head is contained in a PP, while the *wh*-expression is a DP. Case was controlled across all conditions: the verb and preposition selected the same case. The results (Figure 4) show that categorial matching is asymmetric. The strong penalty for PP[DP] and the relative acceptability of DP[PP] are not straightforwardly compatible with a symmetric matching requirement. The results do not support a simple *Shared Structure* analysis based on grafting (van Riemsdijk 2006; Bergsma 2019). Possible explanations include: restrictions on PF deletion of the null head (Bergsma 2023) and independent restrictions on P-stranding in Russian.

Analysis. The empirical data suggests that Russian FR acceptability is gradient rather than categorical. Case hierarchy effects are position-sensitive and emerge in preposed FRs, suggesting the involvement of a post-syntactic process, whereas the position-independent effect of syncretism may point to an earlier stage of the derivation. The gradient nature of case constraints suggests that an *Agree*-based account alone cannot fully capture the observed patterns and is more compatible with an analysis involving PF deletion of the light head (à la Bergsma 2023), potentially supplemented by *Agree*-based feature matching.

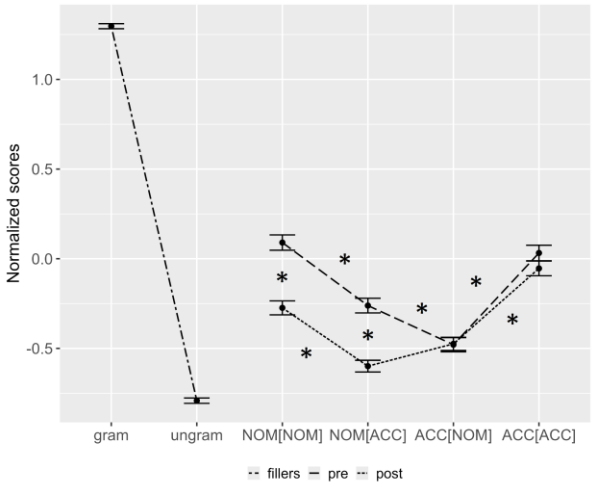


Figure 1. Subject and Object FRs.

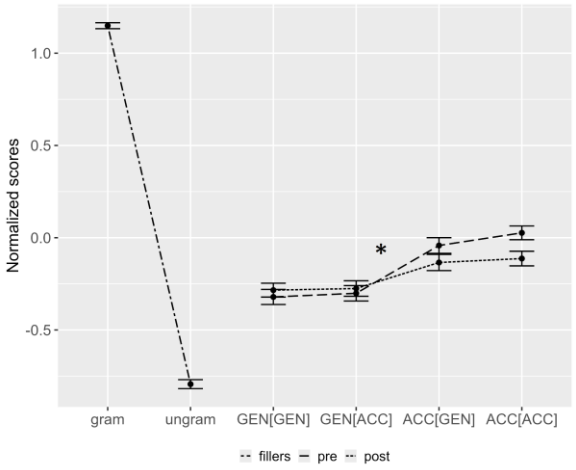


Figure 3. ACC-GEN Syncretism.

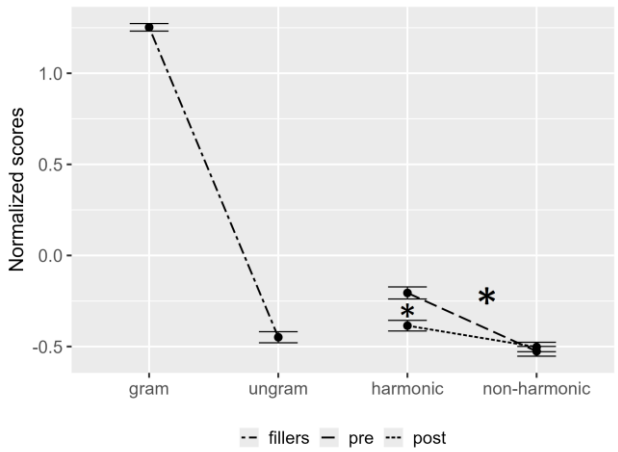


Figure 2. Case Hierarchy.

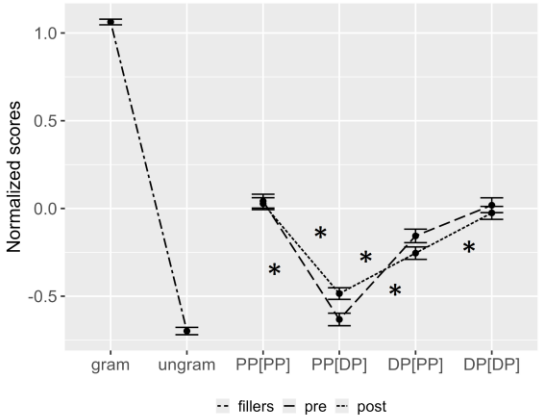


Figure 4. Categorial Matching.

(1) rebënok obnial [ACC kto pozdravil babušk-u
child.NOM hugged who.NOM congratulated granny-ACC

s Roždestvom]
with Christmas

‘The child hugged (the one) who congratulated granny with Christmas.’**ACC[NOM]**

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